

INVX DGO Onboarding Program – Data Lineage

10:00 – 11:00 | 3 Oct 2023





01 Overview of Data Lineage

02 Data Lineage template

03 Data Lineage processes



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WHAT IS DATA LINEAGE? DATA LINEAGE ANSWERS THESE 5 QUESTIONS



DATA SOURCE

Where does the data originate from?



DATA FLOW

How does the data move through the SCBx group?



DATA LOCATION

Where does the data sit? (e.g. zone, dataset)



DATA TRANSFORMATION

What are the changes/transformations applied to the data?

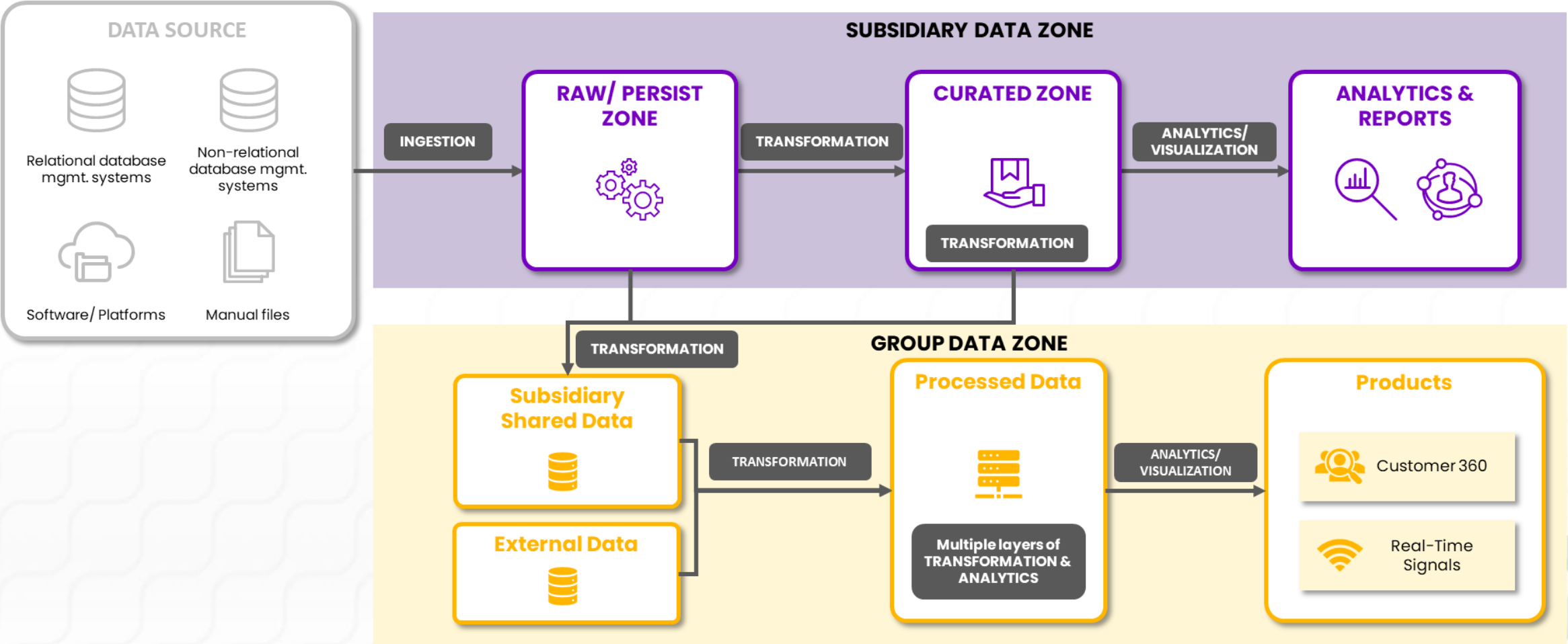


REVISION HISTORY

Who made the changes to the data lineage and when were they made?

AN ILLUSTRATION OF WHAT DATA LINEAGE WILL LOOK LIKE FOR SCBX

Data lineage will be documented end to end – from the point where data lands in the GDZ/ SDZ to when it is consumed for analytics & reports/products.



IN THE TARGET STATE, DATA LINEAGE PROVIDES A DATA MAP TO EXPLAIN AND VISUALIZE ALL THE 5 ELEMENTS

Illustrative

ELEMENTS OF DATA LINEAGE:

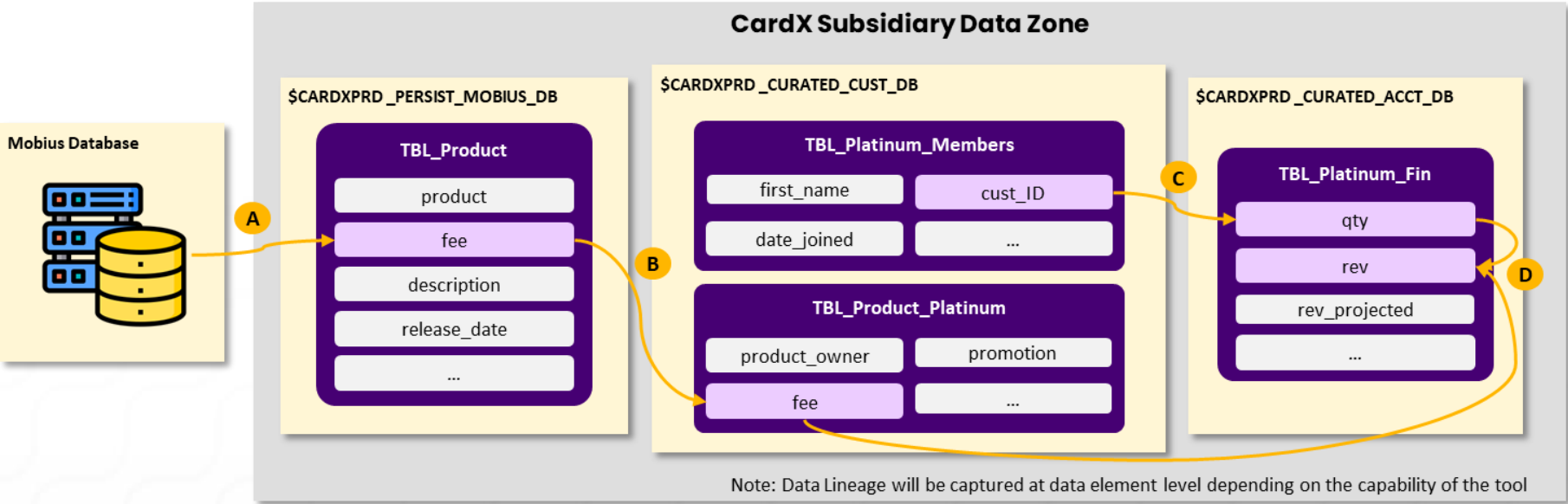
DATA SOURCE

DATA FLOW

DATA LOCATION

DATA TRANSFORMATION

REVISION HISTORY



	Transformation Type	Functional Description	Transformation Details	Updated Date	Updated By
A	Stream ingestion	Ingest data from Mobius database to Persist Zone	= MERGE fee FROM Mobius INTO TBL_Product	20/06/2022	Khun Sam
B	Extraction	Identify annual fee for Platinum product from the list of CardX products	= SELECT fee FROM TBL_Product WHERE Product="Platinum"	20/06/2022	Khun Sam
C	Calculation	Count the number of Platinum product customers	= COUNT(cust_ID) FROM TBL_Platinum_Members	22/06/2022	Khun Ying
D	Calculation	Calculate the revenue from Platinum product (qty * fee)	= SELECT qty * (SELECT fee FROM TBL_Product_Platinum) AS rev	22/06/2022	Khun Ying

CLEAR ROLES & RESPONSIBILITIES DEFINED FOR DATA X AND SUBSIDIARIES TO GOVERN DATA LINEAGE

-  DataX
-  Subsidiaries

Roles from DataX for both GDZ & SDZ:



Metadata Mgmt. Team

Define and review the standards for data lineage for GDZ and SDZ



Metadata Mgmt. Team

Review and monitor data lineage



Technical Data Stewards

Capture data lineage when new data is onboarded, and update data lineage when required



Business Data Steward

Provide support for data lineage capture and update for accountable data



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EXPLANATION OF FIELDS CAPTURED

1	2	3	4	5	6	7	8	9	10
No.	Data Zone	Schema/ Database Name	Data Set Name	Column Name	Transformation Type	Functional Description	Transformation Details	Updated Date	Updated By
4	CardX SDZ	\$CARDXPRD _CURATED_A CCT_DB	TBL_Platinu m_Fin	qty	Calculation	Count the number of Platinum product customers	= COUNT(cust_ID) FROM TBL_Platinum_Members	22/06/2022	Khun Ying

No	Fields	Explanation
1	No.	Running number
2	Data Zone	Name of the data zone that describes where the data element sits (e.g., SDZ, GDZ)
3	Schema/ Database Name	Schema/ Database can be defined as a collection of tables and views of tables – this information includes environment, persist/ curate zone , system name/data subject area
4	Data Set Name	Data set is a collection of information within the schema/ database e.g., table name
5	Column Name	Name of the data column in the data set.
6	Transformation Type	Category of data transformation performed to derive the data element (e.g., calculation, ingestion)
7	Functional Description	Business description of the transformation steps performed to derive the data element
8	Transformation Details	Detailed steps of how the data element was transformed, calculated or processed. <i>Note: Format depends on the tool adopted for data lineage (e.g., SQL script)</i>
9	Updated Date	Date when latest changes were made to data lineage
10	Updated By	User who registered the latest changes to data lineage

*The combination of Data Zone, Schema/ Database Name, Data Set Name, Column Name is distinct and acts as a linkage to technical metadata.



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THESE PROCESSES ARE APPLICABLE TO BOTH THE SUBSIDIARY DATA ZONE AND THE GROUP DATA ZONE



1

Capture Data Lineage Of Newly Onboarded Data*

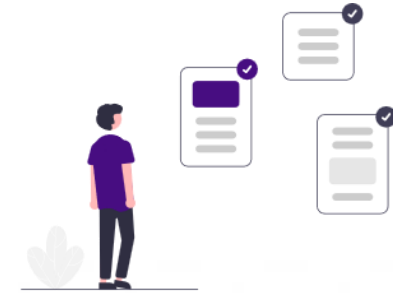
- Technical data steward provides key information for data lineage capture
- Metadata mgmt. officer reviews information provided
- Data owner provides feedback
- Metadata mgmt. officer publishes data lineage



2

Update Data Lineage Captured*

- Technical data steward reviews gap in data lineage and provides update
- Metadata mgmt. officer reviews information provided
- Data owner provides feedback
- Metadata mgmt. officer publishes data lineage



3

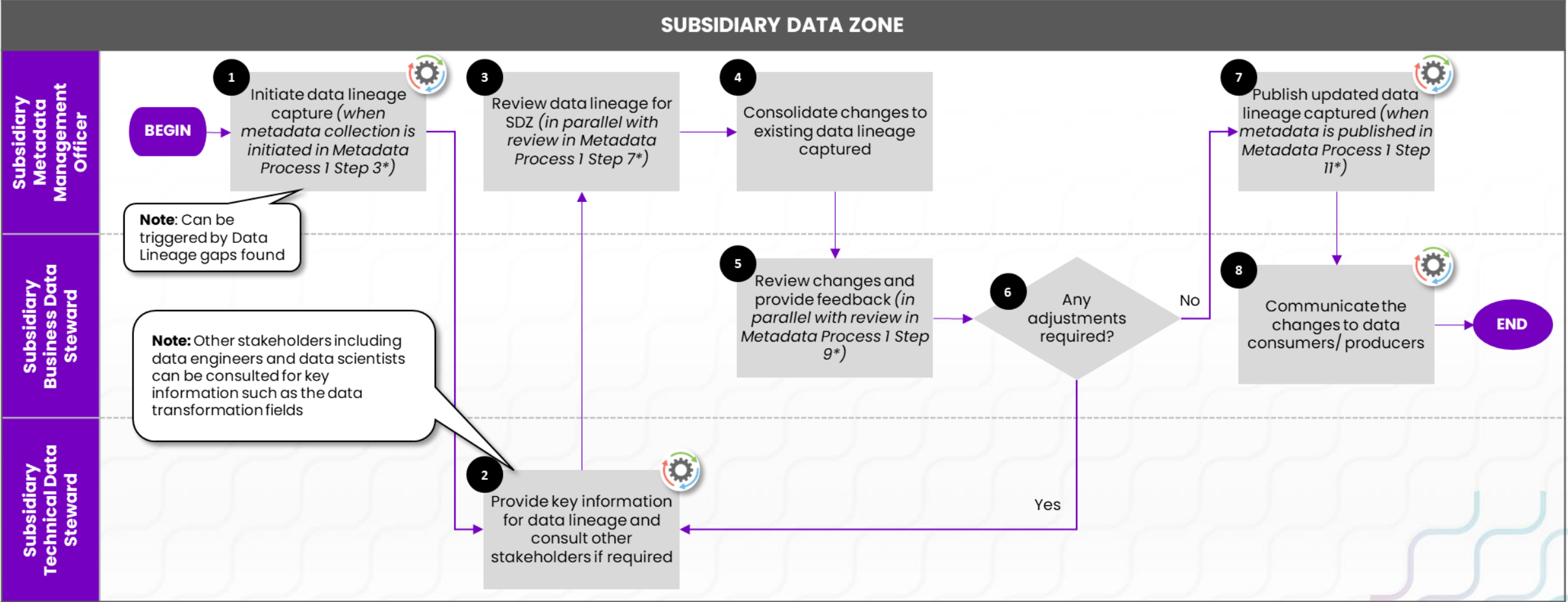
Review Data Lineage Periodically

- DataX metadata mgmt. officer reviews data lineage standards/ processes across GDZ/ SDZ
- DataX and Subsidiary metadata mgmt. officer to inform data consumers/ producers in of changes, if any

**Note: For Data Lineage processes 1 and 2, activities in GDZ will be performed by DataX while activities in SDZ will be performed by Subsidiary*

PROCESS 1 – TO CAPTURE DATA LINEAGE OF NEWLY ONBOARDED DATA

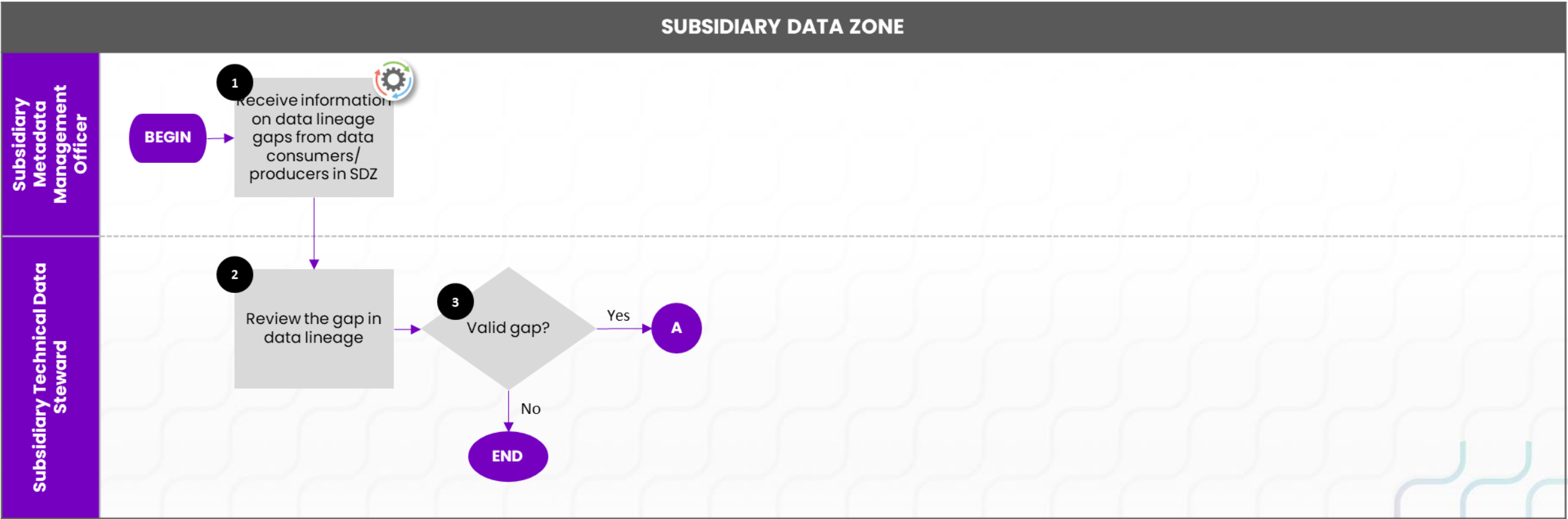
- Prerequisite(s)**
- Technical data steward has been identified by Data Owner after Data Owner receives communication from technical team about new data shortlisted to be onboarded in SDZ from Business Requirement Specification (See Metadata Process 1 Step 2)



PROCESS 2 – TO UPDATE DATA LINEAGE CAPTURED

Prerequisite(s)

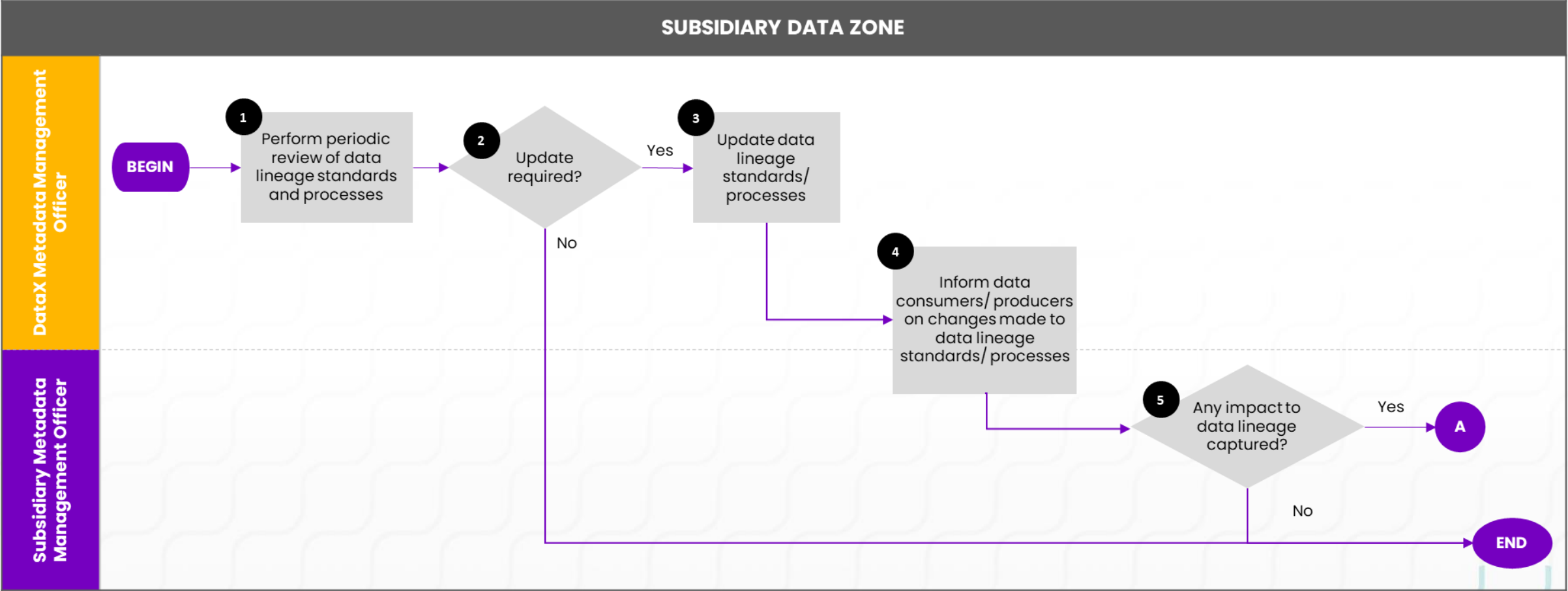
- Metadata management officer receives information about gaps in data lineage in SDZ. This may be triggered by:
 - Data Lineage gaps found by data consumers/ producers
 - Data Lineage gaps found during periodic review in Process 3
 - Updates to any of the 5 data lineage elements – data source, data flow, data location, data transformation, or revision history (e.g. when there is a code change to data pipelines)



A Proceed to Process 1 Step 1

PROCESS 3 – TO REVIEW DATA LINEAGE PERIODICALLY

DataX
Subsidiaries



A Proceed to Process 1 Step 1

Thank You

Please contact DataX Metadata (datax_metadata@data-x.ai)
for more information

